








Advanced Time Series Analysis and Forecasting

Seminar 14: Causal inference for time series



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Acronyms used in this material (1/2)

- **ADH**: Abadie, Diamond and Hainmueller
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CCM**: Convergent Cross Mapping
- **CEE**: Central and Eastern Europe
- **DML**: Double/Debiased Machine Learning
- **EODHD**: EOD Historical Data (market-data provider)
- **ETF**: Exchange-Traded Fund
- **EU**: European Union
- **EU-26**: the 26 EU countries other than Romania
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)

Acronyms used in this material (2/2)

- **HICP**: Harmonised Index of Consumer Prices
- **ITS**: Interrupted Time Series
- **MCI**: Momentary Conditional Independence
- **OECD**: Organisation for Economic Co-operation and Development
- **OLS**: Ordinary Least Squares
- **PC1**: first Principal Component
- **PCMCI**: PC algorithm + Momentary Conditional Independence
- **RMSPE**: Root Mean Squared Prediction Error
- **SC**: Synthetic Control
- **SDID**: Synthetic Difference in Differences
- **TWFE**: Two-Way Fixed Effects
- **UK**: United Kingdom
- **US**: United States
- **USA**: United States of America
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and seminar structure

- Question of the seminar
  - **how do we tell a predictive link from a causal effect in time series, and how do we estimate the effect of a policy on one country?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the Exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	Granger causality from a common driver; a persistent driver	Solved, Proposed	A1
A3, A4	synthetic control weights by hand; placebo p-values	Solved, Proposed	A3
A5, A6	the variance of a cumulative effect; the forbidden comparison	Solved, Proposed	A5
A7, A8	CausalImpact intervals; the bias of a non-orthogonal score	Solved, Proposed	A7
B1, B2	Granger tests: S&P 500, DAX, BET; Bitcoin and Ether	Solved, Proposed	B1
B3, B4	synthetic control: West Germany; robustness for Romania	Solved, Proposed	B3
B5, B6	CausalImpact: Romanian inflation; Bitcoin placebo dates	Solved, Proposed	B5
B7, B8	interrupted time series with a placebo date; PCMCI settings	Solved, Proposed	B7
C1, C2	the Brexit doppelganger on revised data; what is wrong in an AI answer?	Proposed	B3, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations and derivations on paper; Part B: real data, solved in the notebook, each ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
S&P 500, DAX, BET, Bitcoin, Ether	EODHD	daily	2015–2026
HICP and annual inflation, 27 EU countries	Eurostat (2026a)	monthly	2015–2026
GDP per capita, West Germany and OECD	Abadie, Diamond and Hainmueller (2014)	annual	1960–2003
Real GDP, UK and 23 OECD countries	OECD (2026)	quarterly	1995–2019
Weekly realised variances	EODHD; BNR	weekly	2008–2026

- In the notebook: `granger`, `sc_weights`, `sc_demeaned`, `placebo_ratios`, `causal_impact`, `its_design`, `ols_hac`, `pcmci`; no account or key is needed

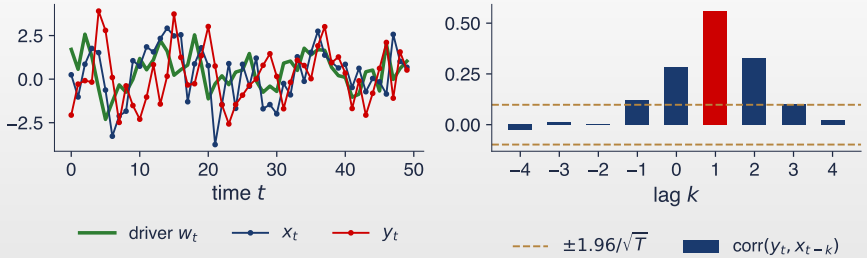
Prerequisites for Today (1/21): Granger causality

- x **Granger-causes** y given z if the lags of x improve the forecast of y_t made from the lags of y and z Granger (1969)

$$y_t = c + \sum_{j=1}^{p} a_j y_{t-j} + \sum_{j=1}^{p} b_j x_{t-j} + \sum_{j=1}^{p} \gamma_j' z_{t-j} + u_t, \qquad H_0 : \; b_1 = \cdots = b_p = 0$$

  - p: the number of lags; b_j: the coefficients of the lags of x; z_t: other conditioning series; u_t: the error
- **Wald test** on the p coefficients b_j, with HAC covariance for returns: Prerequisites (3/21)
  - a large statistic (small p-value): the lags of x help to forecast y
- Granger causality is predictive
  - a common driver, expectations or the timing of observations create it without any effect of x on y

Prerequisites for Today (2/21): a common driver



- Simulated: $w_t = 0.5w_{t-1} + \eta_t$, $x_t = w_{t-1} + e_t$, $y_t = w_{t-2} + u_t$; η_t, e_t, u_t independent $N(0,1)$
  - x has no effect on y: both are noisy readings of the same driver w, x one period earlier
- Right: $\mathrm{corr}(y_t, x_{t-k})$ peaks at $k = 1$ (0.56); with $T = 400$ the Granger statistic of $x \to y$ with $p = 2$ is 164.3
  - Granger causality is real here as prediction and absent as an effect; adding w to z_t would remove it

Prerequisites for Today (3/21): the Wald test and HAC covariance

- **Wald statistic** for H_0: $b = (b_1, \ldots, b_p)' = 0$

$$W = \hat{b}' \left[\widehat{\mathrm{Var}}(\hat{b})\right]^{-1} \hat{b}, \qquad W \sim \chi^2_p \text{ under } H_0$$

  - $\hat{b}$: the OLS estimates of the lag coefficients of x; $\widehat{\mathrm{Var}}(\hat{b})$: their estimated covariance matrix
  - p-value $= \Pr\{\chi^2_p > W\}$; with $p = 2$, $W > 5.99$ rejects at 5%

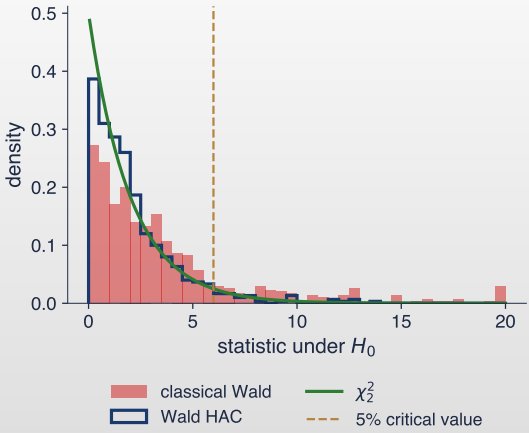
- **Classical** covariance: $\hat{\sigma}^2 (X'X)^{-1}$, valid only for homoskedastic, uncorrelated errors
  - X: the matrix of regressors (constant, lags of y, x, z); $\hat{\sigma}^2$: the residual variance

- **HAC (Newey–West)** Newey and West (1987) covariance, with residuals $\hat{u}_t$ and L lags

$$\widehat{\mathrm{Var}}_{\mathrm{HAC}} = (X'X)^{-1} \Big[\hat{\Gamma}_0 + \sum_{j=1}^{L} \Big(1 - \frac{j}{L+1}\Big)(\hat{\Gamma}_j + \hat{\Gamma}_j')\Big] (X'X)^{-1}, \qquad \hat{\Gamma}_j = \sum_{t=j+1}^{T} \hat{u}_t \hat{u}_{t-j} x_t x_{t-j}'$$

  - x_t: the row t of X; robust to heteroskedasticity (volatility clustering) and to autocorrelation up to lag L

Prerequisites for Today (4/21): size of the classical and HAC Wald tests



- Simulated: y_t and x_t with a common GARCH volatility, independent in the mean (H_0 true)
  - 600 samples of $T = 1000$; Wald test of 2 lags of x in the equation of y
- Rejection rate at 5% (critical value 5.99)
  - classical: 16.3%; HAC with $L = 5$: 4.8%
  - the classical histogram has too much mass to the right of χ^2_2: it finds links that do not exist
- Large past $|x|$ coincide with a large variance of y_t: the classical formula ignores it

Prerequisites for Today (5/21): strength of a Granger link

- **Geweke measure** Geweke (1982): the log ratio of the residual variances without and with the lags of x

$$F = \ln\left(\sigma^2_{\text{restricted}}/\sigma^2_{\text{full}}\right) \geq 0, \qquad TE = F/2$$

  - TE: the transfer entropy for Gaussian data Barnett, Barrett and Seth (2009), in nats: the information that the past of x adds about y_t

- **Population regression** of y on a single regressor x

$$b = \frac{\text{Cov}(y,x)}{\text{Var}(x)}, \qquad \sigma^2_{\text{full}} = \text{Var}(y) - \frac{\text{Cov}(y,x)^2}{\text{Var}(x)}$$

  - without x the residual variance is $\text{Var}(y)$ itself (when the own lags of y do not help)

- **PCMCI** Runge et al. (2019a): a graph-discovery method for many series
  - step PC1: select candidate parents (lagged direct causes) of each variable
  - step MCI: tests each lagged link (next slide); $\tau_{\max}$: the largest lag, α: the significance level

Prerequisites for Today (6/21): PCMCI, convergent cross mapping and volatility proxies

- **Partial correlation** of a and b given a set Z: the correlation of their residuals after regressing both on Z

$$\rho(a, b \mid Z) = \operatorname{corr}\big(a - Z\hat{\beta}_a,\ b - Z\hat{\beta}_b\big), \qquad t = \rho\sqrt{\frac{n-2-|Z|}{1-\rho^2}} \sim t_{n-2-|Z|}$$

  - $|Z|$: the number of conditioning variables; n: the number of observations

- **MCI step** of PCMCI Runge et al. (2019a): the link $X^i_{t-\tau} \to X^j_t$ is kept if

$$\rho\big(X^i_{t-\tau},\ X^j_t \mid \mathcal{P}(X^j_t) \setminus \{X^i_{t-\tau}\},\ \mathcal{P}(X^i_{t-\tau})\big) \neq 0 \text{ at level } \alpha$$

  - $\mathcal{P}(\cdot)$: the parents selected in step PC1; $\tau \leq \tau_{\max}$; a link is directed in time, from the past to the present

- **Convergent cross mapping** Sugihara et al. (2012): x is predicted from the delay vectors of y, $(y_t, y_{t-1}, \ldots, y_{t-E+1})$
  - E: the embedding dimension; skill rising with the length of the library suggests that x drives y (dynamics with deterministic structure)

- **Volatility proxies**: daily $\ln(r_t^2 + 10^{-4})$, the constant avoiding $\ln 0$; weekly $\ln \sum_{t \in \text{week}} r_t^2$

Prerequisites for Today (7/21): synthetic control

- One treated unit ($j = 1$), J donors ($j = 2, \ldots, J+1$), treatment after period T_0 Abadie, Diamond and Hainmueller (2010)

$$\hat{Y}_{1t}(0) = \sum_j w_j Y_{jt}, \qquad w_j \geq 0, \quad \sum_j w_j = 1, \qquad \hat{\tau}_{1t} = Y_{1t} - \hat{Y}_{1t}(0)$$

  - Y_{jt}: outcome of unit j in period t; $\hat{Y}_{1t}(0)$: estimated outcome of the treated unit without the treatment; $\hat{\tau}_{1t}$: the estimated effect

- **Outcome-only SC**: the weights minimise the pre-period fit error

$$\min_w \sum_{t \leq T_0} \left(Y_{1t} - \sum_j w_j Y_{jt} \right)^2$$

Prerequisites for Today (8/21): fit criteria of the synthetic control

- **RMSPE** of the synthetic control over a set of periods $\mathcal{T}$ (pre: $t \le T_0$; post: $t > T_0$)

$$\mathrm{RMSPE}(\mathcal{T}) = \sqrt{\frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}}\Big(Y_{1t} - \sum_j w_j Y_{jt}\Big)^2}$$

  - $|\mathcal{T}|$: the number of periods; a small pre-period RMSPE: the donors reproduce the treated unit before T_0

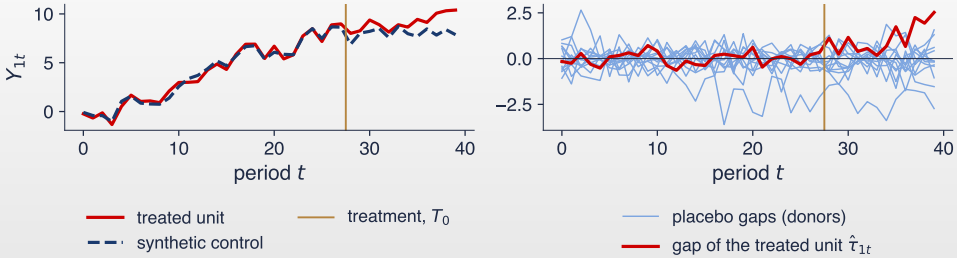
- **ADH predictor matching** Abadie, Diamond and Hainmueller (2015): the weights match a vector of predictors X_1 of the treated unit

$$\hat{w}(V) = \arg\min_{w\ge 0,\ \sum w_j = 1} (X_1 - X_0 w)' V (X_1 - X_0 w)$$

  - X_0: the same predictors for the donors, one column per donor; V: a diagonal matrix of predictor importances
  - V is chosen so that $\hat{w}(V)$ minimises the pre-period RMSPE of the outcome (cross-validation)

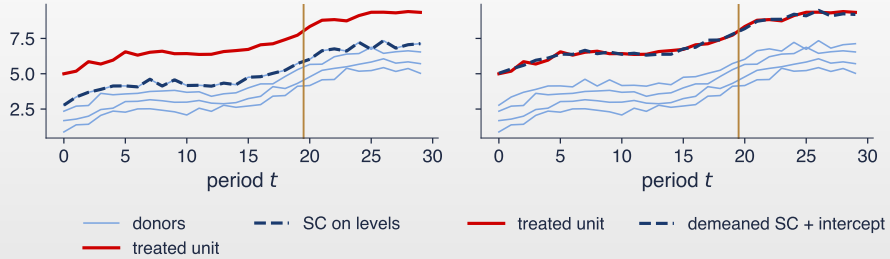
- **Demeaned SC** Ferman and Pinto (2021): $\tilde{Y}_{jt} = Y_{jt} - \bar{Y}_j^{\mathrm{pre}}$ for every unit, the same simplex fit, then $\hat{Y}_{1t}(0) = \bar{Y}_1^{\mathrm{pre}} + \sum_j w_j \tilde{Y}_{jt}$
  - $\bar{Y}_j^{\mathrm{pre}}$: the pre-period mean of unit j; the intercept $\bar{Y}_1^{\mathrm{pre}} - \sum_j w_j \bar{Y}_j^{\mathrm{pre}}$ absorbs a constant level gap

Prerequisites for Today (9/21): a synthetic control and its placebos



- Simulated: 12 donors driven by two common factors; the treated unit is a mix of three donors plus an effect after $T_0 = 28$
  - 6 donors get a weight above 0.02, the largest 0.36; pre-period RMSPE 0.34
- Right: the gap $\hat{\tau}_{1t}$ of the treated unit and the placebo gaps of the donors, each fitted from the others
  - post/pre RMSPE ratio of the treated unit 4.0, the largest of $12 + 1$: p-value $= 1/(12 + 1) = 0.077$

Prerequisites for Today (10/21): outside the convex hull



- Simulated: a treated unit above every donor before T_0; weights $w_j \geq 0$ with $\sum_j w_j = 1$ cannot reach it
  - convex hull: the set of all such weighted averages of the donors; the best fit puts all weight on the highest donor
- Pre-period RMSPE: 2.14 on levels (left), 0.16 after demeaning with an intercept (right)
  - the intercept fits the level gap; the weights fit only the dynamics

Prerequisites for Today (11/21): placebo inference

- **In-space placebos**: treat every donor in turn as if it were the treated unit and compute the ratio

$$r_j = \frac{\mathrm{RMSPE}_{\mathrm{post}}}{\mathrm{RMSPE}_{\mathrm{pre}}}, \qquad \text{p-value} = \frac{\#\{j : r_j \geq r_1\}}{J+1}$$

  - RMSPE: root mean squared prediction error of the synthetic control before (pre) and after (post) T_0; $\#\{\cdot\}$: the number of units
  - r_1: the ratio of the treated unit; the smallest attainable p-value is $1/(J+1)$
- Other placebos
  - in time: a fictitious treatment date in the pre-period
  - leave-one-out: drop each donor with positive weight and refit

Prerequisites for Today (12/21): the 2×2 DiD and the TWFE regression

- **2×2 DiD**: a treated group T and a control group C, one period before and one after

$$\hat{\tau}_{\text{DiD}} = (\bar{Y}_{T,\text{post}} - \bar{Y}_{T,\text{pre}}) - (\bar{Y}_{C,\text{post}} - \bar{Y}_{C,\text{pre}})$$

  - $\bar{Y}_{g,s}$: the mean outcome of group g in period s; the change of C measures what T would have done without treatment
  - identifying assumption: parallel trends of the two groups in the absence of treatment
- **TWFE** (two-way fixed effects) regression on a panel of units i and periods t; staggered adoption: units are treated from different dates

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}$$

  - α_i: unit effects; λ_t: period effects; $D_{it} = 1$ once unit i is treated; $\hat{\beta}$: the TWFE estimate
  - with staggered adoption $\hat{\beta}$ is a weighted average of all 2×2 DiD between groups and periods Goodman-Bacon (2021), including "forbidden" ones that use already-treated units as controls
- **Group-time effects** Callaway and Sant'Anna (2021): one 2×2 DiD per adoption cohort g and period t, using only not-yet-treated or never-treated units as controls

Prerequisites for Today (13/21): interrupted time series and event studies

- **Interrupted time series** Lopez Bernal, Cummins and Gasparrini (2017): fit the pre-period model, extrapolate it after T_0

$$y_t = f(t, \text{season}) + u_t, \ t \leq T_0; \qquad \hat{\tau}_t = y_t - \hat{f}(t), \ t > T_0; \qquad \text{cumulative}: \sum_{t=T_0+1}^{T_0+n} \hat{\tau}_t$$

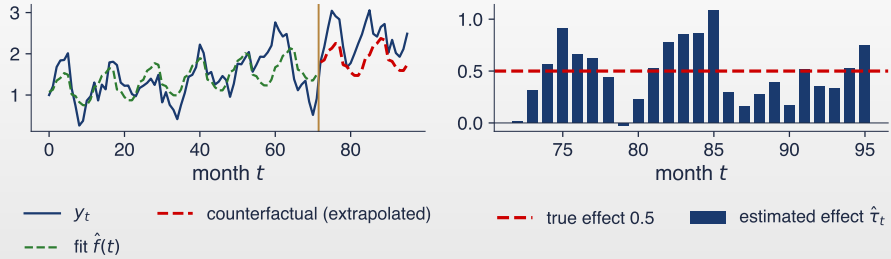
  - f: trend and seasonal effects; $\hat{\tau}_t$: the effect in period t; n: the number of post-intervention periods

- **Event study** MacKinlay (1997): the abnormal return is the return minus what the market model predicts

$$AR_t = r_t - \hat{\alpha} - \hat{\beta} r_t^m, \qquad CAR = \sum_t AR_t$$

  - r_t: return of the asset; r_t^m: market return; $\hat{\alpha}$, $\hat{\beta}$: estimated before the event; the event must be a surprise

Prerequisites for Today (14/21): an interrupted time series

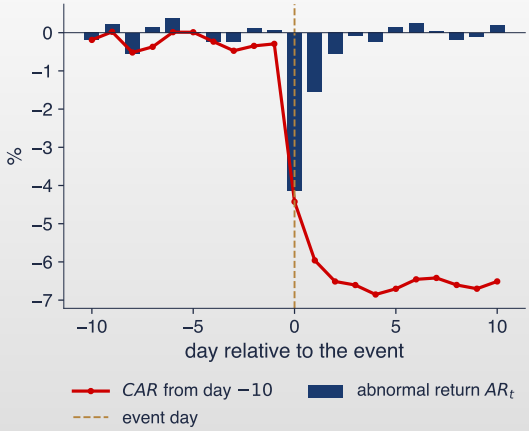


- Simulated monthly series: trend, month effects and AR(1) errors with $\rho = 0.6$; a step of 0.5 after month 72
  - $\hat{f}$ is fitted on the pre-period only and extrapolated; estimated cumulative effect over 24 months 11.6 (true 12.0)
- z-**statistic** of the cumulative effect, with $a = \sum_{t>T_0} x_t$ the sum of the post-period regressors

$$z = \frac{\sum_{t>T_0} \hat{\tau}_t}{\sqrt{n\hat{\Omega} + a' \widehat{\mathrm{Var}}(\hat{\beta}) a}}$$

  - $\hat{\Omega}$: the residual variance (i.i.d.) or the Newey–West long-run variance (HAC); $a' \widehat{\mathrm{Var}}(\hat{\beta}) a$: the error of the extrapolated fit
  - $|z| > 1.96$: significant at 5%

Prerequisites for Today (15/21): abnormal returns around an event



- Simulated: market model estimated on 250 days before the window
  - $\hat{\alpha} = -0.03$, $\hat{\beta} = 1.24$
- The event: a surprise of -4% on day 0 and -1% on day 1
  - $AR_0 = -4.13\%$; CAR over the 21 days $= -6.51\%$
  - before day 0 the AR_t are noise around zero: no anticipation
- Test: $CAR/\left(\hat{\sigma}_{AR}\sqrt{K}\right)$, approximately $N(0,1)$ under no effect
  - K: the number of days in the window; $\hat{\sigma}_{AR}$: the residual standard deviation of the market model

Prerequisites for Today (16/21): dependent errors

- **AR(1) errors**: $u_t = \rho u_{t-1} + \varepsilon_t$, with $|\rho| < 1$ and innovation variance $\sigma^2 = \mathrm{Var}(\varepsilon_t)$

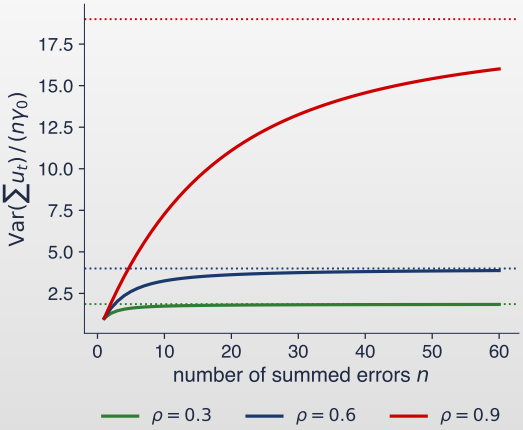
$$\gamma_0 = \frac{\sigma^2}{1-\rho^2}, \qquad \gamma_k = \rho^k \gamma_0$$

  - γ_0: the variance of u_t; γ_k: the autocovariance at lag k
- **Variance of a sum** of n consecutive errors: all autocovariances count

$$\mathrm{Var}\Big(\sum_{t=1}^{n} u_t\Big) = \gamma_0\Big[n + 2\sum_{k=1}^{n-1}(n-k)\rho^k\Big] \approx n\,\Omega, \qquad \Omega = \frac{\sigma^2}{(1-\rho)^2}$$

  - Ω: the long-run variance (Chapter 0); the approximation holds for large n
  - the i.i.d. formula $n\gamma_0$ ignores the autocovariances: with $\rho > 0$ it is too small

Prerequisites for Today (17/21): the variance of a sum of AR(1) errors



- Ratio of the exact variance of $\sum_{t=1}^{n} u_t$ to the i.i.d. value $n\gamma_0$

$$\frac{\mathrm{Var}(\sum u_t)}{n\gamma_0} = 1 + 2\sum_{k=1}^{n-1}\left(1 - \frac{k}{n}\right)\rho^k \;\to\; \frac{1+\rho}{1-\rho}$$

  - dotted lines: the limits $\Omega/\gamma_0 = (1+\rho)/(1-\rho)$
- At $n = 12$: 1.76, 3.38, 8.24 for $\rho = 0.3$, 0.6, 0.9
  - limits 1.86, 4.00, 19.00: the more persistent the errors, the longer the sum takes to reach the limit
- Standard errors scale with the square root of the ratio

Prerequisites for Today (18/21): CausalImpact

- **CausalImpact** Brodersen et al. (2015): a local level plus a regression on control series, fitted on the pre-period

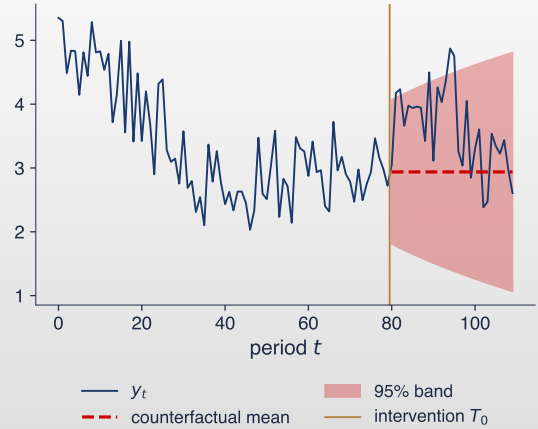
$$y_t = \mu_t + \beta' x_t + \varepsilon_t, \qquad \mu_{t+1} = \mu_t + \eta_t$$

  - μ_t: the local level (Chapter 6); x_t: the control series, with coefficients β; ε_t, η_t: independent noises with variances σ^2_ε, σ^2_η
  - counterfactual paths are simulated after T_0 given the observed controls
- **Forecast uncertainty of the level** h steps after the end T of the fit

$$\mathrm{Var}(\mu_{T+h} \mid \text{data}) = P_T + h\sigma^2_\eta$$

  - P_T: the filtered state variance at T; each step adds the level noise σ^2_η
- Identification: the controls must not be affected by the treatment, and their relation with y must stay stable

Prerequisites for Today (19/21): a local level counterfactual



- Simulated local level, $\sigma_\varepsilon^2 = 0.25$, $\sigma_\eta^2 = 0.02$; Kalman filter on $t < T_0$, then the forecast
  - the counterfactual mean stays at the last filtered level
- Predictive variance of y_{T_0+h}

$$\mathrm{Var}(y_{T_0+h} \mid \text{data}) = P_{T_0} + h\sigma_\eta^2 + \sigma_\varepsilon^2$$

  - here $P_{T_0} = 0.061$: from 0.331 at $h = 1$ to 0.911 at $h = 30$; the 95% band is ± 1.96 standard deviations
- An effect is detected only if it leaves this widening band

Prerequisites for Today (20/21): DML and the Bitcoin case

- **DML** Chernozhukov et al. (2018): the partially linear model

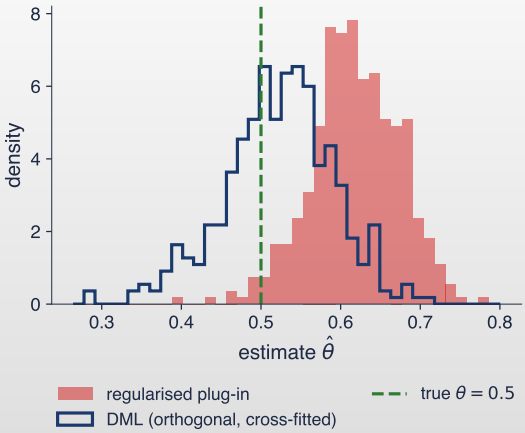
$$y = \theta d + g(X) + u, \qquad d = m(X) + v$$

  - d: the treatment; θ: its effect; X: controls; g, m: unknown functions learned by machine learning; u, v: errors with $\mathbb{E}[u \mid X] = \mathbb{E}[v \mid X] = 0$
- **Orthogonal score**: regress the residual of y on the residual of d

$$\psi = \big(y - \ell(X) - \theta(d - m(X))\big)\big(d - m(X)\big), \qquad \ell(X) = \mathbb{E}[y \mid X]$$

  - first-order errors in $\hat{\ell}$ and $\hat{m}$ do not bias $\hat{\theta}$
- **Bitcoin case**: US spot ETFs approved on 10 January 2024 SEC (2024)
  - outcome: weekly log realised variance, the log of the sum of squared daily returns of the week

Prerequisites for Today (21/21): orthogonal scores and cross-fitting



- **DML estimator** with two folds: learn $\hat{\ell}$, $\hat{m}$ on one half, compute residuals on the other, then swap

$$\hat{\theta} = \frac{\sum_i \left(d_i - \hat{m}(X_i)\right)\left(y_i - \hat{\ell}(X_i)\right)}{\sum_i \left(d_i - \hat{m}(X_i)\right)^2}$$

- Simulated: $n = 500$, 10 controls, ridge learners, 400 samples, $\theta = 0.5$
  - plug-in (ridge of y on d and X): bias 0.118, s.d. 0.056
  - DML: bias 0.022, s.d. 0.068

A1: Granger causality from a common driver [Solved]

Task

- w_t, e_t, u_t independent white noise with variance 1; $x_t = w_{t-1} + e_t$, $y_t = w_{t-2} + u_t$.
- 1. Compute the population coefficient b of the regression of y_t on x_{t-1} and the two residual variances (with and without x_{t-1}; the own lags of y do not help).
- 2. Compute the Geweke measure and the Gaussian transfer entropy.
- 3. What is the effect on y of an intervention that sets x_{t-1} to a new value?
- Report: three numbers; two numbers; one sentence.

Solution

- 1. $\mathrm{Cov}(y_t, x_{t-1}) = \mathrm{Var}(w_{t-2}) = 1$, $\mathrm{Var}(x) = 2$: $b = 0.50$; residual variance 1.50 with x_{t-1}, 2.00 without
- 2. $F = \ln(2.00/1.50) = 0.288$; $TE = F/2 = 0.144$ nats
- 3. Zero: y depends on w only; x is an early, noisy reading of the same driver
- Interpretation: conditioning on w_{t-2} makes $b = 0$; Granger causality depends on the information set

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A2: a persistent common driver [Proposed]

Solution discussed in class

Task

- $w_t = 0.8 w_{t-1} + \eta_t$, $\mathrm{Var}(\eta) = 1$; $x_t = w_{t-1} + e_t$, $\mathrm{Var}(e) = 0.5$; $y_t = w_{t-3} + u_t$, $\mathrm{Var}(u) = 1$. Model: A1.
- 1. Compute $\gamma_0 = \mathrm{Var}(w)$ and $\mathrm{Cov}(y_t, x_{t-1})$.
- 2. Regress y_t on x_{t-1} only: compute b, the residual variance and F against $\mathrm{Var}(y)$.
- 3. Why does persistence of the driver strengthen the spurious link?
- Report: two numbers; three numbers; one sentence.

A3: synthetic control weights by hand [Solved]

Task

- Pre-period outcomes: treated $(3, 4, 5)$; donor A $(2, 3, 4)$; donor B $(6, 7, 8)$. Post-period: treated 9, A 5, B 10.
- 1. Find $w \in [0, 1]$ minimising $\sum_t (Y_{\mathbf{1}t} - wA_t - (1-w)B_t)^{\mathbf{2}}$.
- 2. Compute the synthetic post-period value and the effect.
- 3. Repeat step 1 for a treated unit with pre-period $(9, 10, 11)$; what goes wrong, and how does an intercept help?
- Report: one number; two numbers; two numbers and one sentence.

Solution

- 1. $A - B = (-4, -4, -4)$, $Y_{\mathbf{1}} - B = (-3, -3, -3)$: $w = \frac{(Y_{\mathbf{1}} - B)'(A-B)}{\|A-B\|^{\mathbf{2}}} = 0.75$, an exact fit
- 2. Synthetic 6.25; effect $9 - 6.25 = 2.75$
- 3. The unconstrained optimum is $w < 0$; on the simplex $w = 0$ (all on B) with a residual of 3 in every period (RMSPE 3): the treated unit is outside the hull; demeaned, all three series have slope 1 and any w fits exactly with intercept 3
- Interpretation: Romanian inflation before the 2025 measures (B4) is the same situation: the highest in the EU before the treatment

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A4: placebo p-values [Proposed]

Solution discussed in class

Task

- Post/pre RMSPE ratios of ten units, the treated unit first: 12.0, 4.7, 4.1, 3.3, 3.2, 3.1, 2.9, 2.6, 2.2, 1.9. Model: A3.
- 1. Compute the permutation p-value of the treated unit and of the second unit, if it had been the treated one.
- 2. What is the smallest attainable p-value with 26 donors?
- 3. Why does the test use the ratio and not the post-period RMSPE alone?
- Report: two numbers; one number; one sentence.

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A5: the variance of a cumulative effect [Solved]

Task

- ITS with AR(1) errors, $\rho = 0.5$, $\sigma^2 = 1$; effect summed over $n = 12$ post months.
- 1. Compute γ_0 and the variance of the sum under the (wrong) i.i.d. assumption.
- 2. Compute the exact variance and the long-run approximation $n\Omega$.
- 3. By what factor are the i.i.d. standard errors too small?
- Report: two numbers; two numbers; one number.

Solution

- 1. $\gamma_0 = 1/(1 - 0.25) = 1.333$; i.i.d.: $12\gamma_0 = 16.0$
- 2. Exact: $\gamma_0[12 + 2\sum_{k=1}^{11}(12 - k)0.5^k] = 42.67$; $n\Omega = 12/(1 - 0.5)^2 = 48.0$
- 3. $\sqrt{42.67/16.0} = 1.63$
- Interpretation: with moderate persistence a 95% i.i.d. interval is really an interval of about 77%; B7 shows the consequence on data

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A6: the forbidden comparison [Proposed]

Solution discussed in class

Task

- Two cohorts, periods 0, 1, 2, common trend $+1$ per period. Early cohort treated from period 1 with effects 1 and 3; late cohort treated in period 2 with effect 1. Model: A5.
- 1. Write both outcome paths.
- 2. Compute the DiD of early against late over periods 0–1, and of late against early over periods 1–2.
- 3. Why is the second comparison "forbidden", and what does it do to a TWFE average?
- Report: two paths; two numbers; one sentence.

A7: intervals of a local level counterfactual [Solved]

Task

- Local level with $\sigma_\eta^2 = 0.01$, $\sigma_\varepsilon^2 = 0.94$; filtered state variance at T_0: $P = 0.02$; $n = 25$ post weeks; no controls.
- 1. Compute the predictive variance of y_{T_0+1} and of y_{T_0+25}.
- 2. Show that $\mathrm{Var}(\sum_{h=1}^{n} y_{T_0+h}) = n^2 P + \sigma_\eta^2 \sum_{k=1}^{n} k^2 + n\sigma_\varepsilon^2$ and compute it.
- 3. Compute the half-width of the 95% interval of the average effect.
- Report: two numbers; one derivation and one number; one number.

Solution

- 1. $P + \sigma_\eta^2 + \sigma_\varepsilon^2 = 0.97$; $P + 25\sigma_\eta^2 + \sigma_\varepsilon^2 = 1.21$
- 2. $\sum_h \mu_{T_0+h} = n\mu_{T_0} + \sum_{i=1}^{n}(n-i+1)\eta_i$; the three terms are independent: 91.25
- 3. $1.96\sqrt{91.25}/25 = 0.75$
- Interpretation: an average effect on log variance smaller than about 0.75 cannot be detected: the situation of the Bitcoin ETF analysis (B6)

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A8: the bias of a non-orthogonal score [Proposed]

Solution discussed in class

Task

- $y = \theta d + g(X) + u$; a plug-in estimator solves $\mathbb{E}[(y - \theta d - \hat{g}(X))d] = 0$ with $\hat{g} = g - \delta$; $\mathbb{E}[\delta d] = 0.4$, $\mathbb{E}[d^2] = 2$. Model: A7.
- 1. Derive the probability limit of the plug-in estimator and its bias.
- 2. Show that the orthogonal score $(y - \ell(X) - \theta(d - m(X)))(d - m(X))$ has zero derivative with respect to a perturbation of m at the truth.
- 3. Why does this matter when $\hat{g}$ is a regularised learner?
- Report: one formula and one number; one derivation; one sentence.

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B1: Granger tests on New York, Frankfurt and Bucharest [Solved]

- **Question**: does the S&P 500 Granger-cause the BET, and is this causality?
- **Inputs**: daily log returns of the S&P 500, DAX and BET on common trading days, 2015–2026 ($T = 2\,815$) and 2020–2026; $p = 2$
- **Tasks**
  1. Test S&P 500 $\to$ BET with HAC and with classical covariance, and BET $\to$ S&P 500 with HAC.
  2. Test S&P 500 $\to$ BET conditional on the DAX, and DAX $\to$ BET.
  3. Repeat on 2020–2026.
  4. Interpretation: which explanation fits the pattern of the results?
- **Report**: five Wald statistics with their p-values for each sample, one sentence
- Notebook: section B1

B1: solution [Solved]

Test	2015–2026	2020–2026
S&P 500 → BET, HAC	15.3 (<0.001)	9.3 (0.009)
S&P 500 → BET, classical	34.3 (<0.001)	17.9 (<0.001)
BET → S&P 500, HAC	4.5 (0.105)	2.3 (0.317)
S&P 500 → BET | DAX	9.9 (0.007)	3.1 (0.213)
DAX → BET	1.6 (0.449)	4.2 (0.124)

- Wald statistics (p-values in brackets); the classical statistic is about twice the HAC one: volatility clustering inflates the evidence
- Interpretation: New York closes after Bucharest; the DAX, which closes together with Bucharest, has no lagged effect: the S&P 500 carries news that arrives after the Bucharest close, a timing effect and not a causal effect of US prices

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B2: Granger tests between Bitcoin and Ether [Proposed]

- **Question**: does one crypto-asset lead the other, in returns or in volatility?
- **Inputs**: daily log returns of Bitcoin and Ether (7 days a week), 2021–2026; daily volatility proxy of Prerequisites (6/21); $p = 2$, HAC; model: B1
- **Tasks**
  1. Test Granger causality in both directions for the returns.
  2. Repeat for the volatility proxies.
  3. Report the contemporaneous correlation of the returns.
  4. Interpretation: what does a bidirectional result in volatility suggest about the information set?
- **Report**: four Wald statistics with their p-values, one correlation, one sentence
- Notebook: section B2

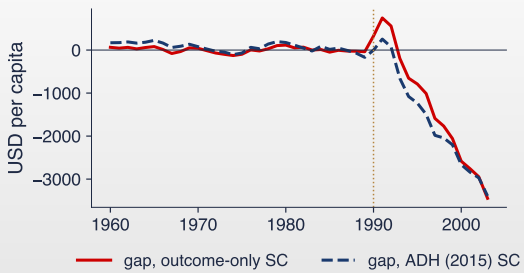
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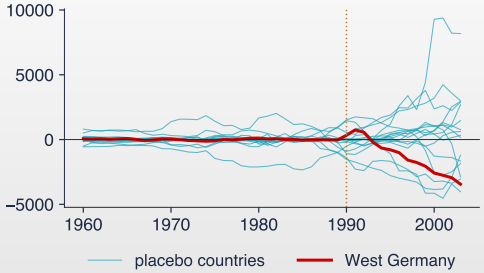
B3: an outcome-only synthetic control for West Germany [Solved]

- **Question**: what changes when the covariates and the cross-validated V of ADH (2015) are replaced by all pre-1990 outcomes?
- **Inputs**: GDP per capita of West Germany and 16 OECD countries, 1960–2003 Abadie, Diamond and Hainmueller (2014)
- **Tasks**
  1. Fit SC weights on the 30 pre-1990 outcomes only; report the main weights and the pre-1990 RMSPE.
  2. Compute the average gap 1990–2003 and compare it with the ADH specification.
  3. Run the in-space placebos for the outcome-only SC and compute the permutation p-value.
  4. Interpretation: which specification would you report, and why?
- **Report**: weights, two RMSPE, two average gaps, one p-value, one sentence
- Notebook: section B3

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B3: solution [Solved]





- Outcome-only weights: USA 0.34; Austria 0.32; Switzerland 0.11; Greece 0.10, …; pre-1990 RMSPE 61 USD against 121 USD for ADH
- Average gap 1990–2003: −1297 USD (outcome-only) against −1588 USD (ADH); placebo $p = 0.059$ (rank 1 of 17)
- Interpretation: a better pre-fit does not guarantee a better counterfactual (Greece gets weight here); the ADH predictors anchor the donors on economic structure; report both, the sign and the significance agree

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B4: robustness of the Romanian synthetic control [Proposed]

- **Question**: is the effect of the 2025 measures on Romanian inflation robust to the donor pool, to single donors and to a placebo date?
- **Inputs**: annual HICP inflation of the 27 EU countries; demeaned SC; pre-period July 2023 – June 2025; effect window July 2025 – June 2026; model: B3
- **Tasks**
  1. Estimate the average gap with three donor pools: EU-26, euro area, non-euro EU.
  2. Drop each donor with weight above 0.02 in turn and re-estimate.
  3. Move the treatment to July 2024 (pre-period July 2022 – June 2024) and estimate the placebo effect over July 2024 – June 2025.
  4. Interpretation: which threat to identification remains after these checks?
- **Report**: three averages, a range, one placebo average, one sentence
- Notebook: section B4

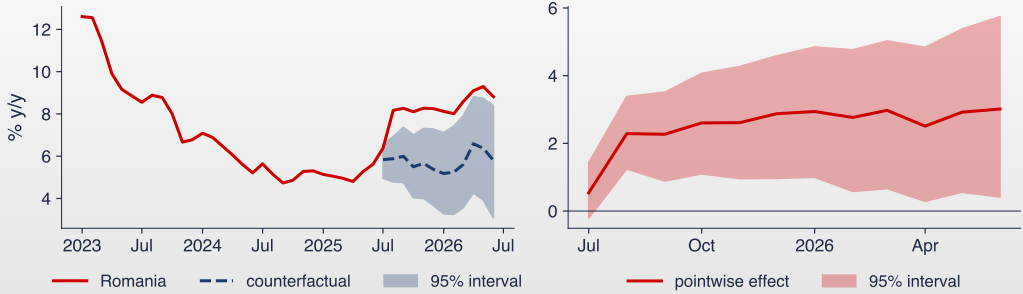
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B5: CausalImpact for Romanian inflation [Solved]

- **Question**: does a state space counterfactual built from CEE inflation rates agree with the synthetic control?
- **Inputs**: annual HICP inflation of Romania; controls: Hungary, Poland, Czechia, Bulgaria, Croatia, Slovakia; pre-period January 2017 – June 2025 (102 months); local level + regression
- **Tasks**
  1. Fit the model on the pre-period and simulate 500 counterfactual paths for July 2025 – June 2026.
  2. Report the average effect with its 95% interval and the largest regression coefficients.
  3. Compare with the demeaned synthetic control estimate with the EU-26 donor pool (2.7 pp; the setting of B4).
  4. Interpretation: which assumption of CausalImpact is most doubtful here?
- **Report**: one effect with interval, three coefficients, one sentence
- Notebook: section B5

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B5: solution [Solved]



- Average effect 2.53 pp, 95% interval [0.80, 4.06]; largest coefficients: Croatia 0.50, Slovakia 0.24, Bulgaria 0.20
- Close to the synthetic control estimates, with a wider interval: the local level absorbs part of the Romania-specific dynamics
- Interpretation: the controls must be unaffected; Bulgaria and Croatia trade with Romania and had their own 2025–2026 measures (Bulgaria joined the euro area in 2026), so the stability of the regression after June 2025 is the doubtful assumption

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B6: placebo dates for the Bitcoin ETF analysis [Proposed]

- **Question**: would a CausalImpact model (as in B5) for the approval of the US spot Bitcoin ETFs (10 January 2024) find effects where there are none?
- **Inputs**: weekly log realised variance of Bitcoin; controls: S&P 500, Nasdaq 100, gold, EUR/USD; data from January 2022 to 7 January 2024; model: B5
- **Tasks**
  1. Run the analysis with fictitious approval dates 2 April, 2 July and 1 October 2023, using only data before the true approval.
  2. Run the true analysis with the S&P 500 as the only control.
  3. Report the four average effects with 95% intervals.
  4. Interpretation: what would it mean if a placebo date gave a significant effect?
- **Report**: four effects with intervals, one sentence
- Notebook: section B6

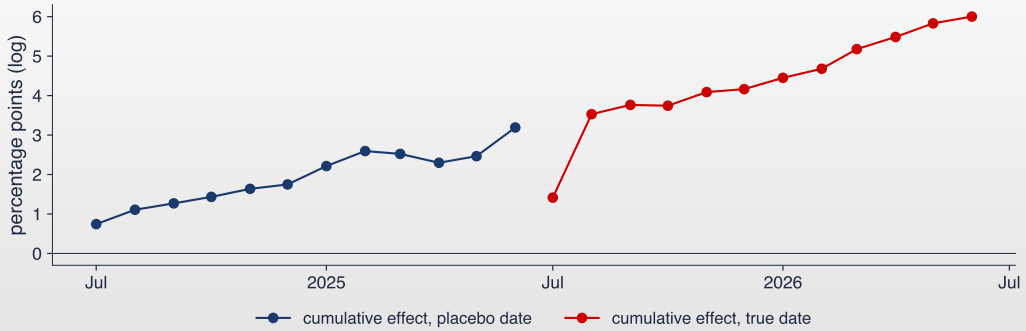
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B7: an interrupted time series with a placebo date [Solved]

- **Question**: does the cumulative effect of July 2025 survive dependent errors, and what does the same design find in July 2024?
- **Inputs**: monthly Romanian HICP inflation (log change), 2015–2026; pre-period model: month effects and a step for July 2021 – June 2023
- **Tasks**
  1. Estimate the 12-month cumulative effect from July 2025 with i.i.d. and with HAC standard errors.
  2. Repeat with a fictitious intervention in July 2024 (model fitted to June 2024).
  3. Compute the z-statistics.
  4. Interpretation: what does the placebo result say about i.i.d. inference?
- **Report**: two effects, four standard errors, four z-statistics, one sentence
- Notebook: section B7

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B7: solution (1/2) [Solved]



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B7: solution (2/2) [Solved]

Intervention	cumulative effect	i.i.d.: se, z	HAC: se, z
July 2025 (true)	6.00	1.55, 3.86	1.98, 3.03
July 2024 (placebo)	3.19	1.59, 2.01	1.99, 1.60

- The true effect stays significant with HAC; the placebo is "significant" at 5% with i.i.d. errors and not with HAC
- Interpretation: i.i.d. inference finds effects at dates where nothing happened; the placebo is also above zero because 2024 inflation stayed above its seasonal norm: the single-series counterfactual is fragile

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B8: PCMCI under different settings [Proposed]

- **Question**: which links of a PCMCI graph of weekly volatilities survive changes of the maximal lag and of the significance level?
- **Inputs**: weekly log realised variances of Nikkei, Euro Stoxx 50, WIG20, BET, S&P 500 and EUR/RON, 2008–2026; PCMCI with partial correlations; model: B7
- **Tasks**
  1. Run PCMCI with $\tau_{\max} = 1, 2, 4$ and $\alpha = 0.01, 0.05$ (six runs).
  2. Count the directed cross links of each run.
  3. List the links found in all six runs.
  4. Interpretation: which links would you report as robust, and with which caveat?
- **Report**: six counts, a list of links, one sentence
- Notebook: section B8

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C1: the Brexit doppelganger on revised data [Proposed]

- **Question**: how fragile is the published Brexit output loss of Born et al. (2019)?
- **Inputs**: OECD quarterly real GDP (1995 = 1) of the UK and the 23 donors, 1995Q1–2019Q4; outcome-only SC; model: B3
- **Tasks**
  1. Write a five-line pre-registration: donor pools, window, outcome, placebos, what counts as a robust loss.
  2. Estimate the gap in 2018Q4 and 2019Q4 with all 23 donors and without Ireland, Iceland, Hungary and Luxembourg.
  3. Run leave-one-out over the donors with weight above 0.02.
  4. Interpretation: turn the results into a project plan (covariates, vintages, conformal inference).
- **Report**: a five-line plan, four gaps, a leave-one-out range, a project plan
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise causal inference for time series. The answer:
- (a) If x Granger-causes y, a policy that changes x will change y.
- (b) With 23 donors, a synthetic control placebo test can reach p = 0.01.
- (c) A perfect pre-treatment fit proves that the synthetic control is unbiased.
- (d) With AR(1) errors and rho = 0.5, i.i.d. standard errors of a long cumulative effect are too small by a factor of about 1.73.
- (e) CausalImpact is valid as long as the control series are correlated with the outcome.
- (f) Convergent cross mapping cannot produce false positives because it uses nonlinear dynamics.
- (g) Two-way fixed effects estimate the average effect on the treated under staggered adoption.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

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Key takeaways

- Granger causality depends on the information set and on timing; test it with HAC and read it as prediction
- Synthetic control is a constrained regression across units; the convex hull and the pre-period length decide whether it can work
- Placebos in space and time are the inference; a placebo date is the cheapest falsification test
- Dependent errors enlarge the uncertainty of cumulative effects; i.i.d. inference finds effects that are not there
- CausalImpact and DML are estimation tools; identification comes from the design

After the seminar

- Lecture 14 develops each topic of today
  - potential outcomes for time series, transfer entropy, PCMCI and CCM, event studies
  - the replications of German reunification and Brexit; augmented SC and SDID; staggered DiD; CausalImpact and DML
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: the Brexit doppelganger across data vintages, pre-registered
- Reading: Abadie, Diamond and Hainmueller (2015); Abadie (2021); Brodersen et al. (2015); Rambachan and Shephard (2021)
- **The seminar is for practice and is not graded**

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